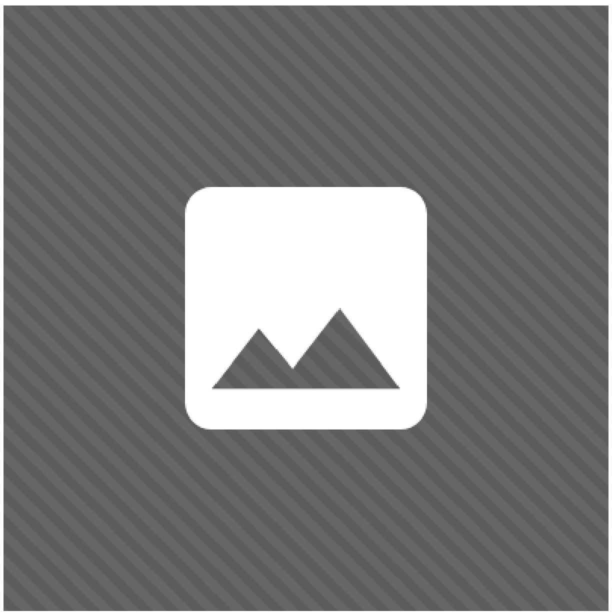
KØBENHAVNS UNIVERSITET
INSTITUT FOR MATEMATISKE FAG



[Specialcouruse NRU]

[Holger Max Fløe Lyng]

[Investigating Methods for Polymicrogyria Classification]

[Undertitel]

Dato: [Juni 2026]

Vejleder: [Navn Navnsen]

Dette er en skabelon for selve opstillingen. Kravene til hvilke oplysninger, der skal medtages, varierer fra fag til fag. Du kan derfor ikke uden videre benytte skabelonen til din egen afhandling. Du får nærmere oplysninger om kravene på dit eget studiekontor!

Institutnavn:	[Institut for Xxxxx]
Afdelingsnavn	[Afdeling for Xxxxx]
Forfatter(e):	[Navn Navnsen]
Titel og evt. undertitel:	[Titel på afhandling, evt. undertitel]
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Indholdsfortegnelse

1 [Timeline]

Lastest submisson day 12.06 - KU I need to research when lastest submisson day at DTU.

First month and one week. Focus on the preprocessing steps, getting a 2D model running with the original setup as in et al guha, optimize setup by splitting training, validation, and test dataset by subjects.

Second month and one week. Focus on the 3D implementation of the preprocessing and model.

Third month and one week. Focus on the missing parts of the project done, and also finish writing the report.

1.1 Project Overview

Submission deadline (KU):	12 June 2026
Submission deadline (DTU):	TBD — confirm early
Project type:	Special course
Public dataset:	PPMR (Kaggle), 15 056 JPEG slices, 23 patients
Clinical dataset:	183 T1w scans, 162 epilepsy patients (110 PMG, 73 non-PMG)
Effective start date:	2 March 2026 (preparation period: 16–28 Feb)
Total working time:	~15 weeks (2 Mar – 12 Jun 2026)

The project will investigate deep learning based detection of PMG from brain MRI. The work will proceed in stages, (1) reproduce the 2D prepressing and classification pipeline of Guha et al. on the public PPMR dataset; (2) improve the methodology, potentially using cDCM from Zhang et al. and the subject level data splitting; and (3) evaluate whether the improved pieline generalises to a clinical epilepsy cohort.

2 [Purpose]

Article 2 - Preprocessing Reproduction (in progress). Reproduction of the preprocessing pipeline described in Article 2 is partially complete. Notably, the paper does not specify how differences in Field of View (FOV) or pixel/voxel spacing are handled across scans. As a first step, we will replicate their network architectures (ResNets and others) to verify whether we can reproduce the overfitting behavior reported in the paper.

3 [Introduction]

This project is an attempt to reproduce Guha et al. [3] paper "Automated detection of polymicrogyria in pediatric patients using deep learning". The article published by scientific reports utilized a publicly available Pediatric polymicrogyria MRI (PPMR) dataset [10] on Kaggle to develop an algorithm for the detection of polymicrogyria (PMG) in epilepsi patients. The PPMR dataset was previously used by the publisher of the dataset for another algorithm for detection of PMG in epilepsi patients [10](maybe remove?). Both attempts lacked domain knowledge necessary to use this dataset correctly, leading to improper data usage and misleading algorithm performance.

PMG is a malformaiton of cortical development characterised by excessive cortical folding and abnormal layering citation. PMG is frequently observed in patients with drug-resistant epilepsy and is commonly associated with neurodevelopmental delay, motor impairments, and cognitive deficiets. Despite its clinical impoartance, PMG remains difficult to characterise reliably due to its heterogeneous redaiological appearance, variable anatomical distribution, and the limited availability of curated neuroimaging datasets.

Previous deep learning studies on PMG have predominantly trained on small, imbalanced pediatric cohorts without rigorous patients-level data splitting citation. Which makes it difficult to distinguish between a genuine good classification performance from artifacts of data leakages, class imbalances, or low-level systematical differences in acquisition, leading to differing FOV and spatial resolution.

In this study, we demonstrate that the high classification accuracy reported by [3] is an artifact of two main methodological failures steming from insufficient domain knowledge of the PPMR daraset. (1) All MRI slices from PMG patients are treated as positive examples, despite the fact that only a subset of slices per patient show PMG related cortical abnormalities; PMG-negative slcies and labeled uncertain slcies – the latter comprising approximately 19% of PMG-patient slices – are incorrectly included as pathological examples. (2) PMG patient images are exported from the picture archibing and communation system (PACS) at approximately 1508×1727, pixels, whereas the healthy control images range from 260×320, pixels to 512×512 pixels, this provides a trivial class-correlated differences that is independent of any cortical pathology. (3) global downsample of dataset size By applied corrective measure to these errors reduces the F1 score of the best performing (ResNet-101) architecture from 0.97 to 0.74, this degradation in performance was observed a across all metrics, excepts accuracy, for the model with corrective measures. These findings are consistent with shortcut learning rahter than any genuine PMG detection.

3.1 text to be placed

Guha and Bhandage's preprocessing pipeline (min-max normalisation, CLAHE, bilateral filtering, Canny edge detection) does not correct for scanner-induced resolution differences; rather, by applying fixed-kernel spatial operations to images with systematically different effective resolutions across classes, it risks amplifying scanner-correlated texture differences into a learnable confound. This may account for the near-perfect classification accuracy reported, which cannot be attributed to PMG pathology alone.

We included 183 T1-weighted MRI scans from 162 unique patients evaluated for epilepsy (77 females; mean ± standard deviation (SD) age at acquisition: 25.8 ± 17.4 years). Of these, 110 scans (49 females; 24.7 ± 15.9 years) were reported to contain radiological evidence of PMG based on clinical radiology reports. The non-PMG group (37 females; 27.5 ± 19.6 years) consisted of scans in which PMG-related terminology appeared in the radiology report, but a subsequent double-rater confirmed the absence of PMG. Of these, 56 cases had other abnormalities and 17 were reported as having a normal MRI. The overall population distribution is shown in Fig. 1(c). MRI scans were acquired in routine clinical practice and retrospectively collected from the Eastern Region of Denmark, encompassing multiple scanner protocols, two magnetic field strengths (B), and three vendors. While no manual quality ratings were available, signal-to-noise ratio (SNR) was computed in white and gray matter as an objective measure of image quality. The mean ± SD SNR was 11.9 ± 2.7 in white matter and 7.9 ± 1.8 in gray matter, indicating that a variability in scan quality could contribute to errors in cortical surface reconstruction and feature extraction.

[1] Severino M, Geraldo AF, Utz N, et al. Definitions and classification of malformations of cortical development: practical guidelines. Brain. 2020;143(10):2874-2894. doi:10.1093/brain/awaa174

[2] Guha S, Bhandage V. Automated detection of polymicrogyria in pediatric patients using deep learning. Scientific Reports. 2025;15(1):41662. doi:10.1038/s41598-025-25572-6

4 Theory

- PMG background bases on `https://www.ncbi.nlm.nih.gov/books/NBK1329/andhttps://medlineplus.gov/genetics/condition/polymicrogyria/#resources`
- Maybe talk alittle about MRI
- Neural networks used. Resnet101 adn DenseNet201
- Maybe preprocessing steps or should that be en methods?

4.1 Polymicrogyria

4.1.1 Neuropathology

Polymicrogyria (PMG) is a malformation of the developing brain characterized by abnormal cortical lamination and an excessive number of abnormally small gyri, resulting in an irregular folding pattern across all or part of the cerebral cortex [9][1]. PMG involves disruption of cortical organisation with or without fusion of the overlying molecular layer, and is frequently characterized by defects in the pial limiting membrane [8]. As a malformation of cortical development (MCD), PMG arises during the post-migrational stage of cortical development, distinguishing it from other MCDs such as lissencephaly, which affects the migration stage[8].

Multiple forms of PMG have been identified. Unilateral PMG affects one hemisphere only, whereas bilateral PMG involves both hemispheres; both have further subcategories. Symptoms depend on how much of the brain is affected: the mildest form, unilateral focal PMG, affects a relatively small area on one side and may cause few or no symptoms, while bilateral forms tends to cause more severe neurological problems[9].

Approximately 20% of all malformations of cortical development are accounted for by PMG, making it one of the most common brain malformations [9].

4.1.2 Aetiology

PMG is aetiologically heterogeneous, arising from both genetic and environmental factors during the period of cortical organisation. More than 40 genes have been implicated in PMG [9, 1]. Among these, mutations in ADGRG1 cause bilateral frontoparietal PMG with cobblestone-like cortical features and cerebellar dysplasia [7].

Infectious causes — most notably congenital cytomegalovirus (cCMV) infection — represent the most common environmental trigger of PMG. The birth prevalence of cCMV is approximately 0.5–1% [2, 4];of these, approximately 10–15% are symptomatic at birth, and among symptomatic cases, approximately 20% develop PMG [5].

Focal intrauterine ischaemia, particularly in monochorionic twin pregnancies with twin-to-twin transfusion, can develop unilateral or focal PMG through disruption of vascular supply to the developing cortex [9, 6]. However, a substantial proportion of PMG cases the aetiology remains unknown despite genetic testing [1].

4.1.3 Clinical Manifestations

Clinical manifestations of PMG depends on the extent and location of the malformation. Epilepsy is the most frequent manifestation and is often resistant to antiseizure medication[8]; onset typically occurs in childhood. Intellectual disability affects the majority of patients with bilateral forms; up to 75% of bilateral perisylvian PMG cases have significant cognitive impairment[9]. and soooo on plsdpll add maybe more

4.1.4 Diagnosis

Magnetic Resonance Imaging (MRI) provides high resolution and adequate contrast to identify the small folds that define PMG, which other imaging modalities lack e.g. Computed tomography (CT).

4.2 MRI

Parameter	Value
Scanner	Siemens Verio, 3 T
Sequence	MPRAGE (T1-weighted)
Repetition time (TR)	1900 ms
Echo time (TE)	2.23 ms
Inversion time (TI)	900 ms
Flip angle	9°
Voxel spacing	$1 \times 1 \times 1$ mm^3

Table 1: Acquisition parameters — Hvidovre Hospital dataset.

Parameter	3 T Siemens Skyra	1.5 T GE Cigna
Sequence	Coronal 3D GRE T1	Coronal 3D GRE T1
Repetition time (TR)	2200 ms	10.44 ms
Inversion time (TI)	1030 ms	450 ms
Echo time (TE)	2.63 ms	4.3 ms
Matrix	320×260	512×512
Slice thickness	1.2 mm	1.2 mm
Field of view (FOV)	20×23 cm	22×27 cm

Table 2: Acquisition parameters — PPMR dataset [10].

4.3 Deep learning for Medical Image Classification

4.3.1 Convolution Neural Networks

Convolution neural networks (CNNs) learn

4.3.2 ResNet-101

Residual learning was introduced by He et al. (2016)[**?**] to address the degradation problem in very deep networks, where accuracy saturates or degrades as depth increases. Each residual block learns a residual mapping $F(x)$ add more... so the block out is $H(x) = F(x) + x$. The skip connection allows gradients to flow directly through the network during backpropagation, making training of networks stable with over 100 layers. ResNet-101, comprising 101 weight layers organised in four residual stages achieved state-of-the-act performance on ImageNet ILSVRC 2015 and has become a widely adopted backbone for medical image and other classification tasks.

4.3.3 DenseNet-201

4.3.4 Transfer learning

5 Methods

5.1 Data

5.1.1 Data splitting

5.1.2 Preprocessing

5.2 Models

5.2.1 ResNet-101 and DenseNet-201

5.2.2 Training Procedure

5.2.3 Evaluation

5.2.4 Ablation study

6 Data

6.1 PPMR dataset

This project utilities the publicly available PPMR dataset introduced by [10] and subsequently used by [3], The dataset consists of coronal 3D gradient-echo T1-weighted MRI slcies from 23 pediatric epilepsy patients with confirmed polymicrogyria and three age. and gender-matched healthy controls per patient, yeilding a 3:1 control-to-case patient ratio.

A critical property of the dataset is that not all MRI slices from PMG patients contrain visible malformation. Each slice was individually labelled by a pediatric neuroradiologist as PMG-positive (label 1), PMG-negative (label 2), or uncertain (label 3) [10]. As shown in Figure 1, only approximately 49% of all PMG-patient slices are labelled as PMG-positive; approximately 30% are PMG-negative, and approximately 19% are labelled uncertain.

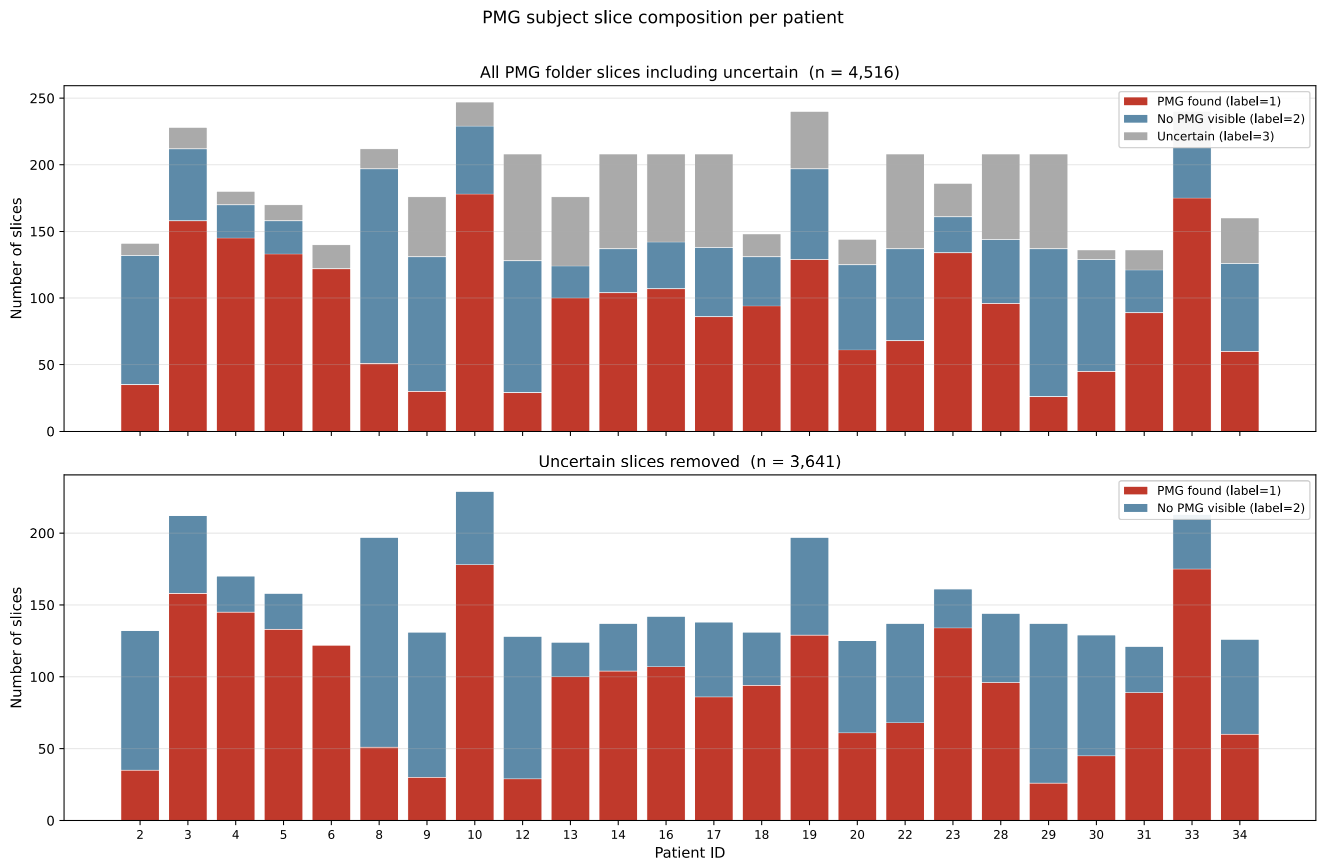

Figure 1: Top: Number of 2D slices per PMG patient labelled as PMG-positive (red), PMG-negative (blue), and uncertain (grey), based on the Kaggle release of the PPMR dataset [10]. Bottom: The same distribution excluding uncertain slices, replicating the figure published in the original dataset paper [10].

Aggregating across all labels, the dataset contains 4,517 attributed to PMG patients and 10,539 attributed to healthy controls. [3] report using precisely these counts in their analysis, treating all 4,517 PMG-patient slices as a single positive class and all 10,539 control slices as a single negative class. Here is the first methodological error: the positive class contains a large proportion of non-pathology slices, which can dilute the PMG-specific attributes and introduce label noise into both training and evaluation. In figure 2 are the full statistics for the dataset.

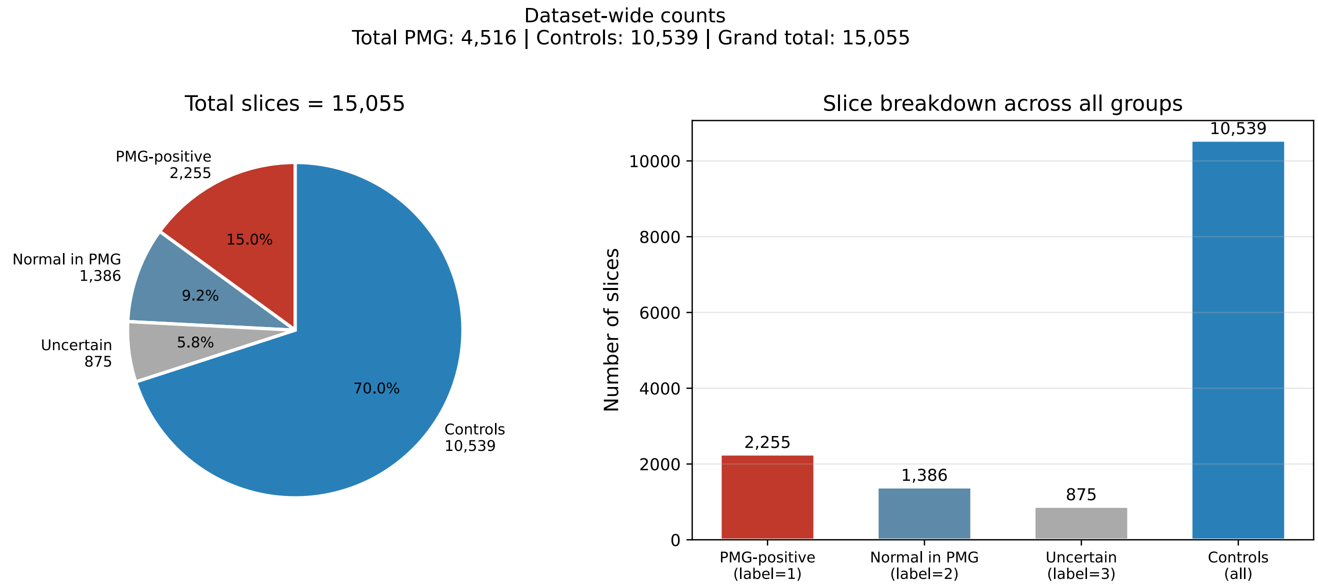

Figure 2: The full statistics of the dataset published on Kaggle[10]. Guha et al. used this data distr

A second confounding factor is the systematic difference in image resolution between classes. As shown in Figure 3, PMG-patient images are exported from the PACS system at approximately 1508×1727 pixels, whereas healthy control images range from 260×320 pixels to 512×512 pixels. When all images are resized to the 224×224 pixel model input, PMG images are compressed by a factor of approximately $6\times$, while HC images are resized by only 1.1–2.3$\times$. Heavy compression averages many original pixels into each output pixel, producing smooth, low-variance texture. Near-native resizing preserves the coarser, higher-variance texture of the original acquisition. A classifier can therefore distinguish the two classes simply by learning that smooth texture implies PMG and coarser texture implies HC – a signal that is present in every patch of every PMG-patient image, regardless of whether cortical pathology is visible in that region.

3.

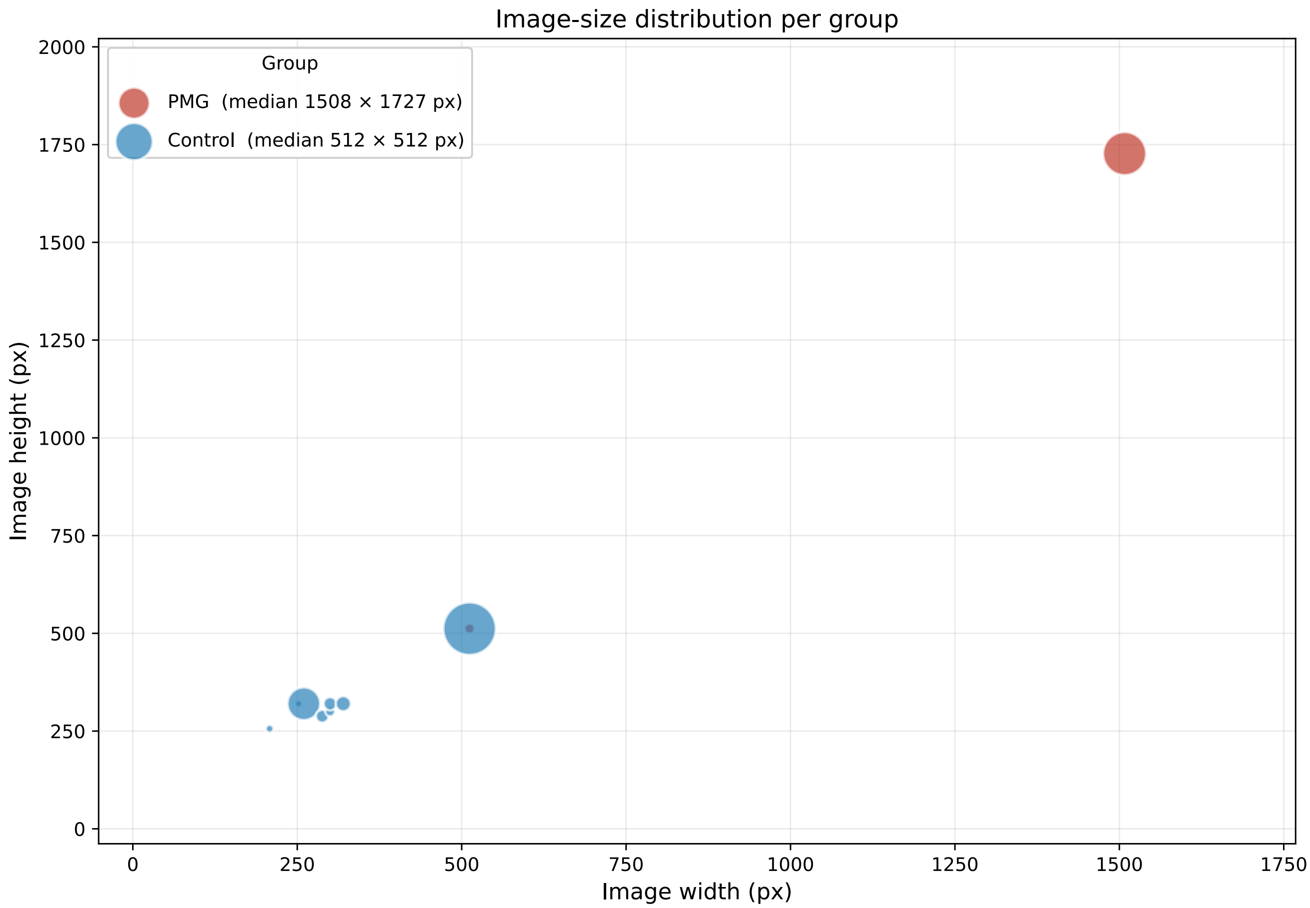

Figure 3: Caption

Guha et al acknowledge the 3:1 natural class imbalance and address it by globally downsampling the 10,539 control slices to 4,517 before splitting the data into training, validation, and test sets. This procedure introduces data leakage. Specifically, the 6,022 discarded control slices may have belonged to patients who would otherwise have been represented in the validation or test set; after downsampling, those patients are absent or underrepresented in evaluation. Consequently, the composition of the test set is determined by the downsampling step and the split step, rather than by the split alone. Furthermore, because balancing is applied globally before the split, the validation and test sets are also balanced at a 1:1 ratio, which is not representative of the true clinical prevalence of PMG. This artificially inflates evaluation metrics — accuracy, recall, and F1 — relative to what would be observed in a real-world deployment where PMG is rare.

A further concern is whether the randomly retained control slices are drawn uniformly across brain positions (anterior, central, posterior) or are disproportionately drawn from the central region, which contains the most anatomically distinctive slices. Position-biased retention could introduce an additional confound, as central slices systematically differ in cortical appearance from peripheral slices regardless of pathology.

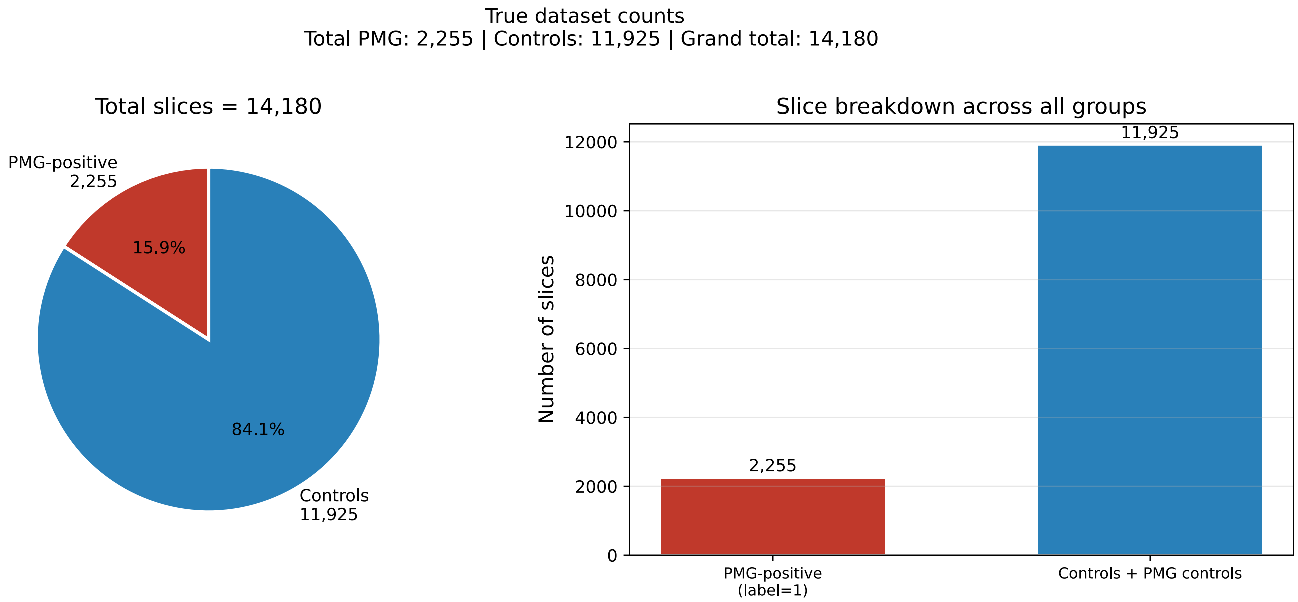
True dataset counts
Total PMG: 2,255 | Controls: 11,925 | Grand total: 14,180
Total slices = 14,180
PMG-positive
2,255
15.9%
84.1%
Controls
11,925
Slice breakdown across all groups
Number of slices
0
2000
4000
6000
8000
10000
12000
2,255
11,925
PMG-positive
(label=1)
Controls + PMG controls

Figure 4:

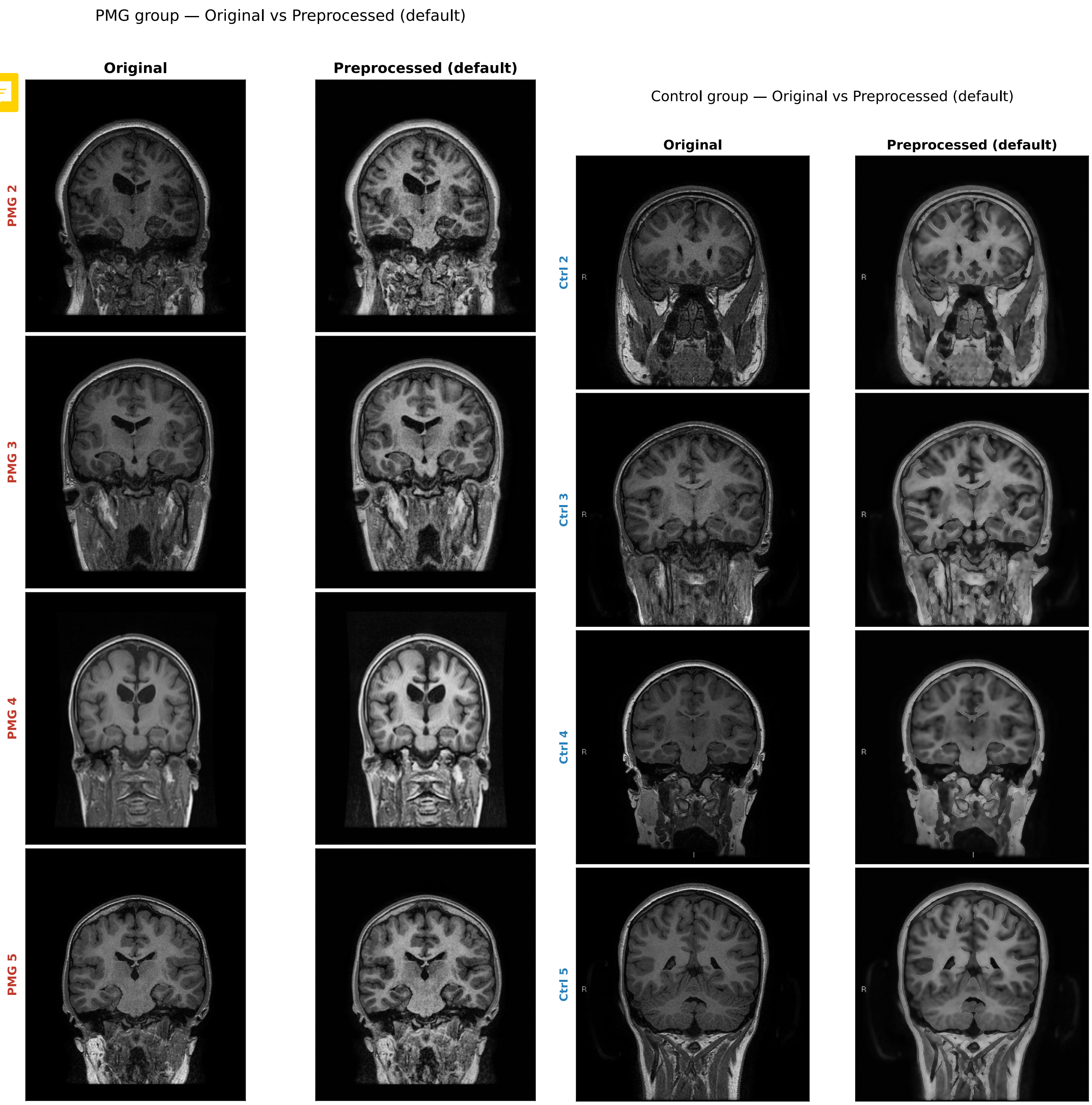

Figure 5: Caption

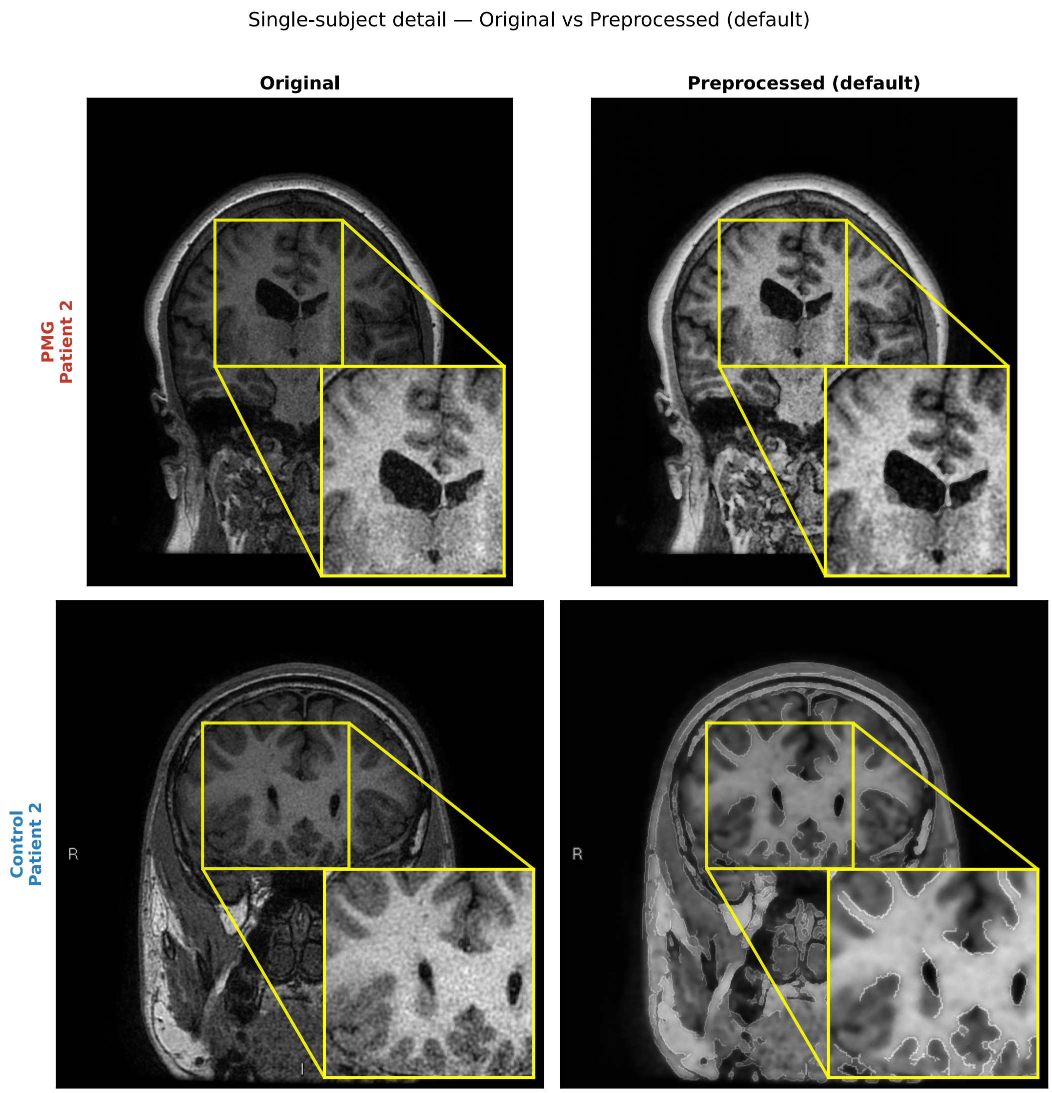

Figure 6: Caption

7 [Problem statement]

7.1 Data problem

Data are split into HC and PMG cases. FOV is different within and between groups of HC and PMG cases. What excatly :::: check FOV, DPI, and JPG size
Cas

7.1.1 Article 1 zheng et al.

Patients under the age of 18 with a diagnosis of PMG were identified from Childrens hospital of Eastern Ontario. A pediatric neuro radiologist with 12 years of experience in reading pediatric

Table 3: Imaging parameters for the coronal T1 weighted sequence.

	Siemens 3T Skyra	**General Electric 1.5T Cigna**
Repetition time (TR)	2200 ms	10.44 ms
Inversion time (TI)	1030 ms	450 ms
Echo time (TE)	2.63 ms	4.3 ms
Matrix	320×260	512×512
Slice thickness	1.2 mm	1.2 mm
FOV	20×23 cm	22×27 cm

brain MRIs. If PMG presence was confirmed, the patient was included in the study. Images of the coronal 3D gradient echo T1 weighted sequence., high resolution sequence and high cortical details. Exported from PACS(Picture archiving and communication system) as JPEG. The cohort consists of 50% females with mean age of 12.3 years at the time of scan, STD 3.6 years, from 5.5 years to 17.8 years.

Dataset consists of 23 patients in total. within each patients MRI, nor all images show evidence of PMG and some slices are normal. Although the ratio between controls and patients is 3:1, the ratio between normal slices and anormals slices is 5:1. Each patients brain includes around 150 scans (full volumes?) on average.

7.1.2 article 2 guha et al.

Data description: Pediatric polymicrogyria MRI (PPMR) dataset by L. Zhang et aladd reff. The dataset consists of 23 patient MRI's, with a combination of PMG cases and healthy control cases.

Ratio between controls and PMG cases is 3:1, ratio between normal cases and anormaly cases is 5:1.

around 150 vrain scans of each patient were included in the dataset?? - Is this true 150 whole scans per subject?

patient cohort has 50 % of the female population?? Mean age of 12.3 during the scan process.

The normal group consists of patients who have shown symptoms of headaches and have undergone MRI scans.

The MRIs where provided in jpeg format from a coronal 3D gradient echo T1 weighted sequence. This dataset contains 15.056 total MRI images (jpeg(slices by slices)), 10.539 are control MRI and 4517 are PMG MRIs.

HMMM???

HC: add details PMG cases: add details

References

[1] Polymicrogyria: MedlinePlus Genetics.

[2] Scott D. Grosse, Danielle S. Ross, and Sheila C. Dollard. Congenital cytomegalovirus (CMV) infection as a cause of permanent bilateral hearing loss: A quantitative assessment. Journal of Clinical Virology, 41(2):57–62, February 2008.

[3] Shagnik Guha, Venkatesh Bhandage, and Aman Agarwal. Automated detection of polymicrogyria in pediatric patients using deep learning. Scientific Reports, 15(1):41662, November 2025.

[4] Aileen Kenneson and Michael J. Cannon. Review and meta-analysis of the epidemiology of congenital cytomegalovirus (CMV) infection. Reviews in Medical Virology, 17(4):253–276, 2007. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/rmv.535.

[5] George Lawson, Alexander Sheeka, Pritika Gaur, Styliani Alifieraki, Nigel Basheer, Wajanat Jan, Carolina Kachramanoglou, and Hermione Lyall. Polymicrogyria in infants with symptomatic congenital cytomegalovirus at birth is associated with epilepsy: A retrospective, descriptive cohort study. Developmental Medicine and Child Neurology, 67(8):1026–1033, August 2025.

[6] Kaylee B. Park, Teresa Chapman, Kimberly A. Aldinger, Ghayda M. Mirzaa, Jordan Zeiger, Anita Beck, Ian A. Glass, Robert F. Hevner, Anna C. Jansen, Desiree A. Marshall, Renske Oegema, Elena Parrini, Russell P. Saneto, Cynthia J. Curry, Judith G. Hall, Renzo Guerrini, Richard J. Leventer, and William B. Dobyns. The spectrum of brain malformations and disruptions in twins. American Journal of Medical Genetics Part A, 185(9):2690–2718, 2021. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/ajmg.a.61972.

[7] Xianhua Piao, R. Sean Hill, Adria Bodell, Bernard S. Chang, Lina Basel-Vanagaite, Rachel Straussberg, William B. Dobyns, Bassam Qasrawi, Robin M. Winter, A. Micheil Innes, Thomas Voit, M. Elizabeth Ross, Jacques L. Michaud, Jean-Claude Déscarie, A. James Barkovich, and Christopher A. Walsh. G protein-coupled receptor-dependent development of human frontal cortex. Science, 303(5666):2033–2036, March 2004.

[8] Mariasavina Severino, Ana Filipa Geraldo, Norbert Utz, Domenico Tortora, Ivana Pogledic, Wlodzimierz Klonowski, Fabio Triulzi, Filippo Arrigoni, Kshitij Mankad, Richard J Leventer, Grazia M S Mancini, James A Barkovich, Maarten H Lequin, Andrea Rossi, and on behalf of the European Network on Brain Malformations (Neuro-MIG). Definitions and classification of malformations of cortical development: practical guidelines. Brain, 143(10):2874–2894, October 2020.

[9] Chloe A. Stutterd, William B. Dobyns, Anna Jansen, Ghayda Mirzaa, and Richard J. Leventer. Polymicrogyria Overview – RETIRED CHAPTER, FOR HISTORICAL REFERENCE ONLY. In Margaret P. Adam, Sarah Bick, Ghayda M. Mirzaa, Roberta A. Pagon, Stephanie E. Wallace, and Anne Amemiya, editors, GeneReviews®. University of Washington, Seattle, Seattle (WA), 1993.

[10] Lingfeng Zhang, Nishard Abdeen, and Jochen Lang. A novel center-based deep contrastive metric learning method for the detection of polymicrogyria in pediatric brain MRI. Computerized Medical Imaging and Graphics, 114:102373, June 2024.